

Digital Electronics

Transistors

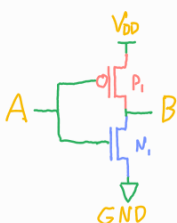


If $g=0$, d and s are disconnected
If $g=1$, d and s are connected



If $g=0$, d and s are connected
If $g=1$, d and s are disconnected

Not gate



Logic Levels

Since voltage is never exactly equal to V_{DD} or 0 , logic levels are used to determine whether a voltage should be read as a 0 or a 1 .

V_{OH} and V_{IH} are the minimum voltages for output and input of a gate to be high respectively, the difference between them being the high noise margin NM_H . Similarly, V_{OL} and V_{IL} are the maximum voltages for low output and input, with NM_L the noise margin.

Static Discipline - if inputs meet valid input thresholds, the system guarantees that the outputs also meet valid output thresholds.

Boolean Algebra

A **minterm** is the **AND** of all inputs to a function, each appearing once (either as itself or negated)

A **maxterm** is the **OR** of all inputs to a function, each appearing once (either as itself or negated)

Any Boolean expression can be written as **minterms ORed together** - **Sum of Products** - and as **maxterms ANDed together** - **product of Sums**.

Example Truth Table

$F(A, B, C)$

Sum of Products:

OR together the combinations of the variables that output 1

$$F(A, B, C) = \bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot B \cdot C + A \cdot \bar{B} \cdot \bar{C} + A \cdot B \cdot \bar{C}$$

Product of Sums:

AND together the maxterms made up of the negations of the variables that output 0.

$$F(A, B, C) = (A+B+\bar{C}) \cdot (A+\bar{B}+C) \cdot (\bar{A}+B+\bar{C}) \cdot (\bar{A}+\bar{B}+C)$$

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Karnaugh Maps

Used for simplifying Boolean expressions

Example Truth Table

$F(A, B, C)$

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

AB	00	01	11	10
0	1	0	1	1
1	0	1	0	0

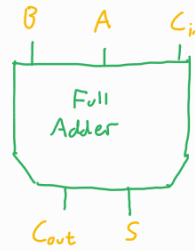
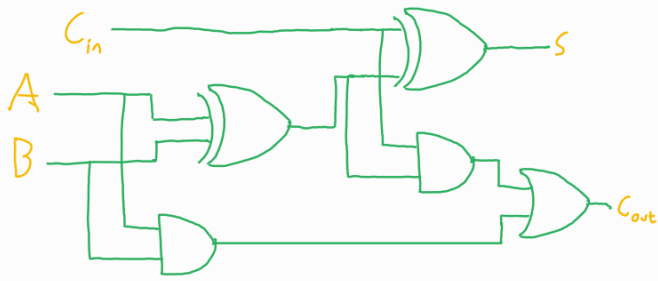
$$F(A, B, C) = \bar{A} \cdot B \cdot C + \bar{B} \cdot \bar{C} + A \cdot \bar{C} \quad (\text{SOP})$$

AB	00	01	11	10
0	1	0	1	1
1	0	1	0	0

$$F(A, B, C) = (A+\bar{B}+C) \cdot (\bar{A}+\bar{C}) \cdot (B+\bar{C}) \quad (\text{POS})$$

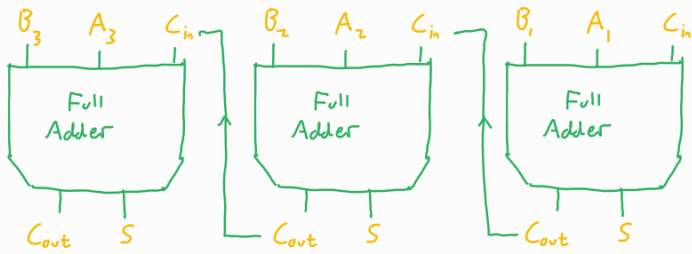
Circuits

Adder: Adds two bits (plus an input carry bit)



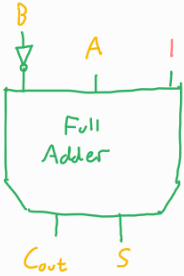
Adders can be chained so the carry bit from the first addition is input as C_{in} to the next one:

3 bit adder

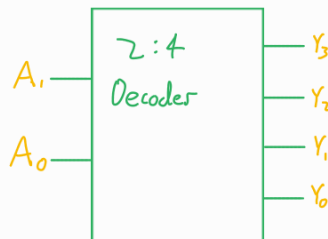
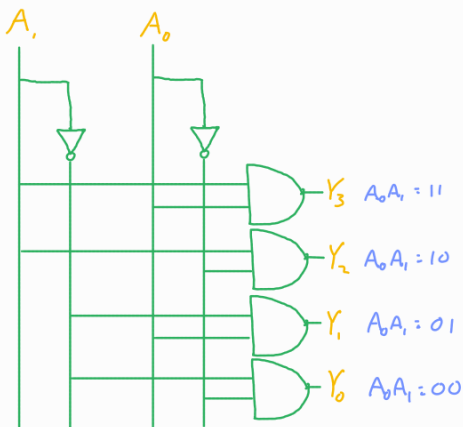


Subtractor: Subtracts one binary number from another.

Take an adder and negate the B input and set the initial C_{in} to 1
→ This is equivalent to two's complement-ifying B .

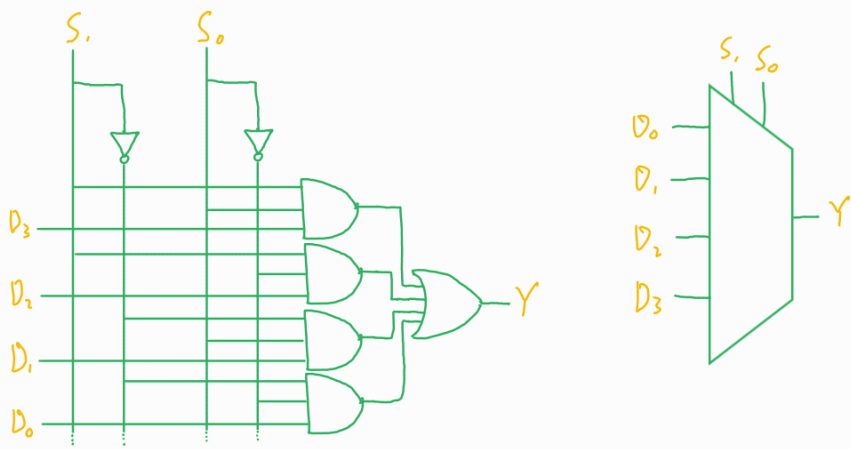


Decoder: n inputs, 2^n outputs. Only one output is 1, which one depends on input.



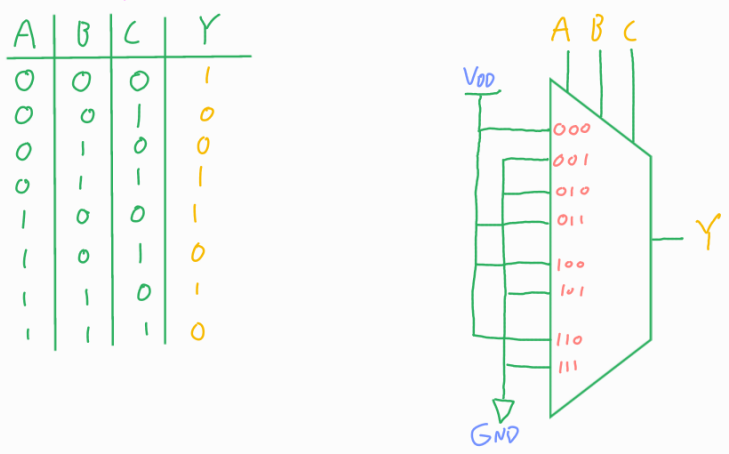
Multiplexor (Mux): $k + 2^k$ inputs

— First k inputs S are the selector. The value of the selector determines which of the other 2^k inputs that the single output takes the value of.

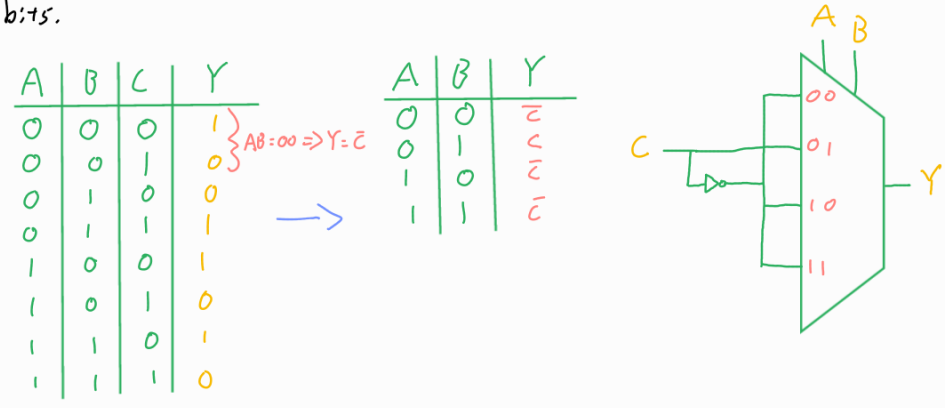


Multiplexers can also be used to implement Boolean functions

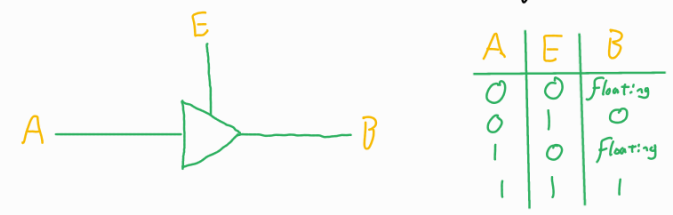
Example



We can use a smaller mux by using one of our inputs A, B, C in the back of the mux, rather than using all 3 as selector bits.



Tristate Buffers: The input signal passes through a tristate buffer only if the enable signal is 1. Otherwise, its output is 'floating', i.e. has no effect on any other signals



Inverted TSBs also exist which invert the signal if $E=1$, otherwise 'floating'.

